

Ong Shi Hui, Magenta

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EDUCATION

Singapore University of Technology and Design (SUTD) **2023 – 2027**

B.Eng. in Computer Science and Design with *Software Engineering Focus*

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Elements of Software Construction, Databases, Network Security, Computer Systems Engineering

WORK EXPERIENCE

People's Association | Software Engineer **Sept 2025 – Present**

- Volunteered with Mountbatten CIT under the Sparks Division to deliver community-focused software solutions (see CompostKaki project).
- Applied agile collaboration with local stakeholders to meet deployment requirements, integrating Next.js & Supabase

GenEd | CoFounder, Software Engineer **Mar 2025 – Present**

- Led full-stack development of AI-powered learning platform using React, Node.js, PostgreSQL, Express.js, Clerk.
- Collaborated with the Malaysian government to build a mentor chatbot for students using Twilio APIs & LLMs.
- Awarded the S\$6,000 Baby Shark Fund by SUTD Venture Innovation and Entrepreneurship.

CPF Board | Automation, Software Engineer **Sept 2025 – Dec 2025**

- Developed automation web application solutions to streamline workflows, reduce manual effort, and improve operational efficiency across departments.
- Built back-end web scraping and data processing pipelines using JavaScript to extract staff information from staff directory, reducing runtime from 40s to 1.6s per department (25x faster).
- Designed maintainable logic systems and modular automation flows, integrating business requirements into scalable and production ready software solutions; collaborated in Agile sprints with Git/CI/CD.

PROJECTS

CompostKaki — Community Composting Platform **Jun 2025 – Present**

- Built and deployed a full-stack CRUD platform using Next.js, TailwindCSS and Supabase (auth + database) to track compost bins, assign tasks, and log community efforts.
- Collaborated with Mountbatten CIT leaders to deploy a solution that supports community sustainability goals; Awarded SGD \$20k support by the People's Association.

BefriendersCircle — Fullstack Caregiver Platform **May 2025 – Aug 2025**

- Developed scalable backend architecture (Express.js, MongoDB) and modular frontend (React) with JWT authentication, RESTful APIs and protected routing.
- Ensured reliability through modular components, well-structured API routes, and robustness/equivalence testing with Jest, Supertest, and Cypress.

TaskTales — Gamified Task Management App **Jan 2025 – Apr 2025**

- Created Android app (Java, Firebase) with OOP design, authentication, and interactive dashboards.
- Awarded S\$900 Singtel Information Systems & Programming Award

LepakLah! — AI Senior-Centric Mobile App **Jun 2024 – Sep 2024**

- Developed a Flutter-based app with a Buddy Matching Algorithm for personalized connections, to improve accessibility to activity centers for seniors. Deployed on RedHat OpenShift, Docker, Kubernetes; Won 3rd Place at Dell InnovateFest 2024.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL — **Frameworks & Tools:** React, Next.js, Node.js, PostgreSQL, MongoDB, Firebase, Clerk, Jest, Cypress — **Data & ML:** Pandas, NumPy, Matplotlib, Scikit-Learn — **Deployment:** AWS (basic), Docker, Kubernetes, OpenShift — **Development:** CI/CD, Git, Agile